Cell membranes — Solution

Y22 (2022) — English translation

A1 — 1.00 pts

Write down the differential equation that the electric field potential φ must satisfy. Hint: according to the Boltzmann distribution, the concentration n of particles with energy E is proportional to $\exp\left(-\frac{E}{kT}\right)$, where T is the absolute temperature of the medium.

Let us consider a section of the solution outside the membrane, which has a unit area and is enclosed between the coordinates x and $x + \mathrm{d}x$. For it, by the Gauss theorem

$$E(x + \mathrm{d}x) - E(x) = \frac{1}{\varepsilon\varepsilon_0} en(x)\mathrm{d}x.$$

Since

$$E(x) = -\frac{\mathrm{d}\varphi}{\mathrm{d}x},$$

we obtain

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = -\frac{e}{\varepsilon\varepsilon_0} n(x).$$

The potential energy of a positive ion in this field is equal to $e\varphi(x)$, therefore for the concentration of ions from the Boltzmann distribution we obtain

$$n(x) = n_0 e^{-\frac{e\varphi(x)}{kT}}.$$

Finally,

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = -\frac{en_0}{\varepsilon\varepsilon_0} \exp\left(-\frac{e\varphi}{kT}\right).$$

Answer:

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = -\frac{en_0}{\varepsilon\varepsilon_0} \exp\left(-\frac{e\varphi}{kT}\right)$$

A2 — 0.30 pts

Write down the boundary conditions that are satisfied by the potential φ on the surface of the membrane, i.e. find $\varphi(0)$, $\frac{d\varphi}{dx}(0)$.

By condition $\varphi(0) = 0$. Since the cell membrane is a uniformly charged sphere, the field outside at its surface is

$$E(0) = \frac{-\sigma}{\varepsilon\varepsilon_0},$$

from

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x}(0) = -E(0) = \frac{\sigma}{\varepsilon\varepsilon_0}.$$

Answer:

$$\varphi(0) = 0, \quad \frac{\mathrm{d}\varphi}{\mathrm{d}x}(0) = \frac{\sigma}{\varepsilon\varepsilon_0}$$

A3 — 0.80 pts

Identify n_0, A_1 and B_1. Express $\varphi(x)$ and $n(x)$ (the concentration of positive ions at point x) in terms of σ, ε, ε_0, T, e and k.

Let's substitute the function $\varphi(x) = A_1 \ln(1 + B_1 x)$ into the equation obtained in **A1**:

$$-\frac{A_1 B_1^2}{(1+B_1 x)^2} = -\frac{e n_0}{\varepsilon\varepsilon_0}(1+B_1 x)^{-\frac{eA_1}{kT}}$$

From here we get

$$A_1 = \frac{2kT}{e}, \quad B_1 = \sqrt{\frac{n_0 e^2}{2\varepsilon\varepsilon_0 kT}}.$$

Substituting $\varphi(x) = A_1 \ln(1 + B_1 x)$ with the found A_1 and B_1 into the equation $\frac{\mathrm{d}\varphi}{\mathrm{d}x}(0) = \frac{\sigma}{\varepsilon\varepsilon_0}$ obtained in point **A2**, we have

$$A_1 B_1 = \frac{\sigma}{\varepsilon\varepsilon_0},$$

from which we directly it turns out

$$B_1 = \frac{\sigma e}{2\varepsilon\varepsilon_0 kT}, \quad n_0 = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}.$$

Then the expression for the potential takes the form

$$\varphi(x) = \frac{2kT}{e}\ln\left(1 + \frac{\sigma e x}{2\varepsilon\varepsilon_0 kT}\right).$$

From the equation

$$n(x) = n_0 e^{-\frac{e\varphi(x)}{kT}}$$

we finally obtain

$$n(x) = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}\frac{1}{\left(1 + \frac{\sigma e x}{2\varepsilon\varepsilon_0 kT}\right)^2}.$$

Answer:

$$A_1 = \frac{2kT}{e}, \quad B_1 = \frac{\sigma e}{2\varepsilon\varepsilon_0 kT}, \quad n_0 = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}, \varphi(x) = \frac{2kT}{e}\ln\left(1 + \frac{\sigma e x}{2\varepsilon\varepsilon_0 kT}\right), n(x) = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}\frac{1}{\left(1 + \frac{\sigma e x}{2\varepsilon\varepsilon_0 kT}\right)^2}$$

A4 — 0.40 pts

Show that the resulting $n(x)$ guarantees the electrical neutrality of the system.

Answer: Indeed, the charge of positive ions per unit area of the membrane is equal to:

$$e\int_0^{+\infty} n(x)\,\mathrm{d}x = \frac{\sigma^2 e}{2\varepsilon\varepsilon_0 kT}\int_0^{+\infty}\frac{1}{\left(1 + \frac{\sigma e x}{2\varepsilon\varepsilon_0 kT}\right)^2}\,\mathrm{d}x = \sigma\int_0^{+\infty}\frac{1}{(1+y)^2}\,\mathrm{d}y = \sigma$$

A5 — 1.50 pts

Per unit area of the membrane, find the energy of the electric field U_f and the sum of the potential energies of positive ions U_e in this field.

Electric field energy density

$$u_f = \frac{1}{2}\varepsilon\varepsilon_0 E^2 = \frac{1}{2}\varepsilon\varepsilon_0\left(\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right)^2.$$

From the formulas obtained in the previous paragraphs we have

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x}=\frac{2kT}{e}\frac{\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}}{1+\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}x}.$$

From here we find the field energy:

$$U_f=\frac{1}{2}\varepsilon\varepsilon_0\int_0^{\infty}\left(\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right)^2\mathrm{d}x=\frac{\sigma kT}{e}\int_0^{\infty}\frac{\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}}{\left[1+\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}x\right]^2}\mathrm{d}x==\frac{\sigma kT}{e}\int_0^{\infty}\frac{1}{(1+y)^2}\mathrm{d}y=\frac{\sigma kT}{e}.$$

The sum of the potential energies of positive ions in this field is

$$U_e=\int_0^{\infty}en(x)\varphi(x)\mathrm{d}x=\frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}2kT\int_0^{\infty}\frac{\ln\left(1+\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}x\right)}{\left(1+\frac{\sigma e}{2\varepsilon\varepsilon_0 kT}x\right)^2}\mathrm{d}x==\frac{2\sigma kT}{e}\int_0^{\infty}\frac{\ln(1+y)}{(1+y)^2}\mathrm{d}y=-\frac{2\sigma kT}{e}\int_0^{\infty}\ln(1+y)\mathrm{d}\frac{1}{1+}$$

Answer:

$$U_f=\frac{\sigma kT}{e},\quad U_e=\frac{2\sigma kT}{e}$$

B1 **0.80 pts**

Identify A_2 and B_2. Express your answer in terms of T, ε, ε_0, e, k and n_0.

Similar to **A1**, the potential satisfies the differential equation

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2}=-\frac{en_0}{\varepsilon\varepsilon_0}\exp\left(-\frac{e\varphi}{kT}\right)\quad\Big[-D<x<D,\ \varphi(0)=0\Big].$$

Substituting the function $\varphi(x)=A_2\ln[\cos(B_2x)]$ into it, we have

$$\frac{A_2B_2^2}{\cos^2(Bx)}=\frac{en_0}{\varepsilon\varepsilon_0}\big[\cos(Bx)\big]^{-\frac{eA}{kT}}.$$

From here we immediately obtain

$$A_2=\frac{2kT}{e},\quad B_2=\sqrt{\frac{n_0e^2}{2\varepsilon\varepsilon_0 kT}}.$$

Answer:

$$A_2=\frac{2kT}{e},\quad B_2=\sqrt{\frac{n_0e^2}{2\varepsilon\varepsilon_0 kT}}$$

B2 **0.50 pts**

Let's introduce the value $\theta=B_2D$. From the boundary conditions on the membrane surface, obtain an equation relating θ and e, σ, D, ε, ε_0, k and T.

Similar to **A2**, the boundary condition has the form

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x}(-D)=\frac{\sigma}{\varepsilon\varepsilon_0}.$$

For the function $\varphi(x)=A_2\ln[\cos(B_2x)]$ we have

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x}=-A_2B_2\,\mathrm{tg}(B_2x),$$

from where

$$A_2B_2\,\mathrm{tg}(B_2D)=\frac{\sigma}{\varepsilon\varepsilon_0}.$$

Substituting the expression for A_2 from **B1**, we obtain

$$B_2 D \operatorname{tg}(B_2 D) = \frac{\sigma e D}{2\varepsilon\varepsilon_0 kT},$$

which can be rewritten through $\theta \equiv B_2 D$ in the form

$$\theta \operatorname{tg}\theta = \frac{\sigma e D}{2\varepsilon\varepsilon_0 kT} \quad \left[0 < \theta < \frac{\pi}{2}\right].$$

Answer:

$$\theta \operatorname{tg}\theta = \frac{e\sigma D}{2\varepsilon\varepsilon_0 kT} \quad \left[0 < \theta < \frac{\pi}{2}\right]$$

B3 — 0.40 pts

Express n_0 in terms of D, σ, ε, ε_0, T, e, k and θ.

Since $B_2 = \frac{\theta}{D}$, then

$$n_0 = \frac{2\varepsilon\varepsilon_0 kT}{e^2} B_2^2 = \frac{2\varepsilon\varepsilon_0 kT\theta^2}{e^2 D^2}.$$

Answer:

$$n_0 = \frac{2\varepsilon\varepsilon_0 kT\theta^2}{e^2 D^2}$$

B4 — 0.50 pts

Find the difference $\Delta n \equiv n(\pm D) - n_0$ between the concentrations of positive ions at the surface of the two membranes and in the middle between them. Express your answer in terms of σ, ε, T and k.

From the Boltzmann distribution, when substituting the function $\varphi(x) = A_2 \ln[\cos(B_2 x)]$, we obtain an expression for concentration in the following form:

$$n(x) = n_0 e^{-\frac{e\varphi(x)}{kT}} = n_0 [\cos(B_2 x)]^{-\frac{eA_2}{kT}} = n_0 \frac{1}{\cos^2 B_2 x},$$

from

$$n(\pm D) = n_0 \frac{1}{\cos^2(B_2 D)} = n_0 \left(1 + \operatorname{tg}^2 B_2 D\right) = n_0 + n_0 \operatorname{tg}^2\theta.$$

Note that

$$n_0 \operatorname{tg}^2\theta = \frac{2\varepsilon\varepsilon_0 kT}{D^2 e^2}\left(\theta \operatorname{tg}\theta\right)^2 = \frac{2\varepsilon\varepsilon_0 kT}{D^2 e^2}\left(\frac{\sigma e D}{2\varepsilon\varepsilon_0 kT}\right)^2 = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}.$$

Thus,

$$\Delta n = n(\pm D) - n_0 = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}.$$

Answer:

$$\Delta n = \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}$$

B5 — 0.50 pts

Show that the resulting $n(x)$ guarantees the electrical neutrality of the system.

Answer: Similar to **A4**, the charge of positive ions per unit area of the system is equal to:

$$\int_{-D}^{D} en(x)\,\mathrm{d}x = en_0 \int_{-D}^{D} \frac{\mathrm{d}x}{\cos^2(B_2 x)} = \frac{2e}{B_2}\frac{2\varepsilon\varepsilon_0 kT\theta^2}{e^2 D^2}\tan\theta = \frac{2e}{B_2}\frac{2\varepsilon\varepsilon_0 kT\theta}{e^2 D^2}\frac{\sigma e D}{2\varepsilon\varepsilon_0 kT} = 2\sigma$$

B6 **1.50 pts**

Find the total force f acting per unit area of the membrane. Hydrostatic pressure can be neglected. Express your answer in terms of e, D, θ, ε, ε_0, k and T Hint: positive ions in water behave like an ideal gas and are locally in thermal equilibrium.

Each membrane is subject to the charges of positive ions and the other membrane. Since the system as a whole is electrically neutral, the total charge of the ions and one of the membranes is equal to the charge of the membrane, taken with the opposite sign. Thus, the system is in some sense equivalent to a flat capacitor, for which, as is known,

$$f_e = \frac{\sigma^2}{2\varepsilon\varepsilon_0},$$

and f_e is directed from one membrane to the other. The pressure created by positive ions as an ideal gas is

$$f_h = n(\pm D)kT = \left(n_0 + \frac{\sigma^2}{2\varepsilon\varepsilon_0 kT}\right)kT = \frac{2\varepsilon\varepsilon_0 k^2T^2\theta^2}{e^2D^2} + \frac{\sigma^2}{2\varepsilon\varepsilon_0}.$$

The resulting force will push the membranes apart and is equal to

$$f = f_h - f_e = n_0 kT = \frac{2\varepsilon\varepsilon_0 k^2T^2\theta^2}{e^2D^2}.$$

Second method The $x = 0$ plane divides the system into two parts. Due to the symmetry and electrical neutrality of the system as a whole, each of these parts will also be electrically neutral, and therefore these parts will not interact with each other electrostatically. Thus, the interaction of the parts is reduced only to the pressure of an ideal gas of positive ions, equal to

$$f_{h0} = n_0 kT.$$

Since each of the parts is in equilibrium, then

$$f = f_{h0} = n_0 kT = \frac{2\varepsilon\varepsilon_0 k^2T^2\theta^2}{e^2D^2}.$$

Answer:

$$f = \frac{2\varepsilon\varepsilon_0 k^2T^2\theta^2}{e^2D^2}$$

B7 **1.80 pts**

For relatively small distances D the dependence $f(D)$ has the form: $f = k_1D^\alpha$, and for large ones: $f = k_2D^\beta$. Find α and β.

As was obtained in **B6**,

$$f \propto \left(\frac{\theta}{D}\right)^2,$$

where θ, according to **B2**, satisfies the equation

$$\theta \operatorname{tg}\theta = \frac{e\sigma D}{2\varepsilon\varepsilon_0 kT},$$

i.e. θ also depends on D. Consider the case $\frac{e\sigma D}{2\varepsilon\varepsilon_0 kT} \ll 1$. For him $\operatorname{tg}\theta \approx \theta$, therefore

$$\theta \approx \sqrt{\frac{e\sigma D}{2\varepsilon\varepsilon_0 kT}} \propto \sqrt{D},$$

because

$$f \propto \left(\frac{\sqrt{D}}{D}\right)^2 = \frac{1}{D},$$

i.e. $\alpha = -1$. Let us now consider the case $\frac{e\sigma D}{2\varepsilon\varepsilon_0 kT} \gg 1$. For him $\theta \approx \frac{\pi}{2} = \mathrm{const}$, therefore

$$f \propto \frac{1}{D^2},$$

i.e. $\beta = -2$.

Answer:

$$\alpha = -1, \quad \beta = -2$$